

Project Brief (BLUF)

Attached: Project Pitch (Project2_Pitch.pdf) -- submitted at Checkpoint 1. The project scope, dataset, business problem, and stakeholder analysis remain unchanged.

Scope Update

The project has been completed through Phase 6 (Final Evaluation) with no changes to the original scope or direction. All four planned models (Logistic Regression, Random Forest, XGBoost, SVM) were trained and compared via 5-fold stratified cross-validation. After two-stage hyperparameter tuning (RandomizedSearchCV followed by GridSearchCV), **XGBoost was selected as the final model** based on the highest tuned AUC (0.703 CV, 0.693 test) and recall (0.638 CV, 0.621 test). **Both success targets -- AUC > 0.65 and Recall > 0.50 -- were met on the held-out test set.** SVM was excluded from tuning due to a 45-minute training time that made hyperparameter search infeasible, while delivering the worst baseline AUC (0.675).

The most notable finding was that XGBoost started as the worst-performing baseline (AUC 0.679) but became the best model after tuning -- a +0.024 improvement, the largest of any model. SHAP analysis revealed that length of stay, prior inpatient visits, and discharge disposition were the top three predictors, while the HbA1c measurement feature that showed a strong univariate association in Phase 2 did not survive multivariate modeling. The business impact simulation showed positive net benefit at every threshold tested, with the optimal operating point (threshold 0.30) yielding approximately **\$4.6 million in annual net benefit** for a 10,000-discharge hospital after accounting for \$500/patient intervention costs against \$13,000/readmission penalty savings.

Metric	CV	Test	Target	Status
ROC-AUC	0.703	0.693	> 0.65	MET
Recall	0.638	0.621	> 0.50	MET
Precision	0.124	0.118	Monitor	--
F1-Score	0.208	0.199	Maximize	--

CV-to-test drop was less than 0.01 on AUC, confirming strong generalization with no meaningful overfitting.